# Correct-Response Novelty in Full-Weight LLM Evolution


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## Abstract

We test an adaptation of historical response novelty to full-weight Gaussian evolution strategies on witnessed arithmetic problems. The comparison uses one paired training seed and an untrained base model. The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference. Problem-resampling intervals exclude training-seed variability. All tables are generated from persisted evaluations; incomplete outcomes are labeled.


## Context, motivation and introduction

The experiment asks whether exploration among correct answer forms improves held-out task success. ExTra [1] supplies the historical semantic novelty concept; ES at Scale [2] supplies the full-weight optimization reference. Their combination uses population-level ES rather than GRPO, with a different sibling population and task, so it is an adaptation rather than a reproduction. A reward-only ES control is necessary because ES can itself change solution coverage [3].

## Background and related work

Novelty-seeking ES was previously studied in control [4]. LLM-specific alternatives include EVOL-RL's within-group novelty [5], DARLING's quality/diversity objective [6], and the model/task coevolution of AC/DC [7]. These use different training or population mechanisms. The repository literature audit distinguishes prompt evolution, artifact search and actual model-parameter training, with code/license verification.

## Method

The model is Qwen/Qwen2.5-0.5B-Instruct, revision 7ae557604adf67be50417f59c2c2f167def9a775. Each generation evaluates 30 independent Gaussian candidates (sigma 0.001) on 32 rotating training problems and 16 fixed training probes. Population-z-scored task reward is combined with 1.0 times correctness-gated cosine novelty for the main condition. The archive retains 6 correct embeddings per probe after seeded insertion. These coefficients weight noise directly, without another normalization; effective alpha is 0.0005. FP32 masters reconstruct BF16 candidates afresh and tied tensors are updated once. Prefix regeneration and backpropagation are absent.

Qwen/Qwen2.5-0.5B-Instruct: 24 layers, width 896, 14 query heads and 2 key/value heads; 494,032,768 unique parameter elements in 290 tensors. Architecture and counts come from the pinned pretrained files and the generation-zero checkpoint audit.

For candidate i, Q_i is the mean response reward: exact success receives 1, a legal expression using every supplied number but missing the target receives 0.1, and an invalid expression receives 0. For probe p, n_i,p = I(correct_i,p) [1 - max cosine(e_i,p, z)]. The reference set consists of correct historical responses and correct responses from other candidates on that same probe; self-comparisons are excluded. An empty set has maximum zero. N_i is the probe mean. History is frozen while scoring the whole generation, then correct embeddings are appended in seeded shuffled order and the oldest entries are removed.

Z_i = (Q_i - mean(Q)) / (std_population(Q) + 1e-8), with zero Z for constant Q. W_i = Z_i + 1.0 N_i for the main arm and W_i = Z_i for control. The update is theta_next = theta + (alpha / population) sum_i W_i epsilon_i. Each epsilon_i is independent standard Gaussian noise over every unique parameter tensor. A common positive novelty bonus is retained. Alpha includes the inverse-noise factor; it is not an additional alpha/sigma update. MiniLM is frozen, and both conditions compute the same categories of probe work.

Protocol amendment A replaced combined-reward ranks with source-aligned z-scored quality plus post-normalization novelty. It was published after calibration and partial original-model evaluation were visible, but before either trained condition began. Both pilot arms started from the original pretrained weights using the amended rule from their first generation. The original registration, changed configuration, code revisions and unchanged absolute deadline are retained in the run record. This ordering follows ES at Scale Algorithm 2 and ExTra Equation 2; it is not evidence that the hybrid outperforms the abandoned rule.

## Registered longer continuation

This follow-up continues each arm's own generation-19 FP32 checkpoint to a fixed target of 100 generations, preserving the archive, RNGs and global schedules. The duration is the only training change. The original pilot was already observed and published. The new primary endpoint is the fresh 2048-problem set, excluding all earlier number multisets. The original test is secondary. Parent checkpoints receive fresh-only evaluation after continuation training, so the change contrast pairs four outcomes per problem. The earlier execution was interrupted after main generation 100 and control generation 31, with 19 candidates saved in the next control population. Its deadline (2026-11-08T18:50:55.807504+00:00) expired during the outage; that attempt remains incomplete. The separately registered recovery grants at most 10 hours, including 7 for unfinished control training and 3 for final work, capped by the original whole-week deadline. This amends the pretrained-stage allocation and may displace later studies. The main checkpoint is reused unchanged. Every non-metadata checkpoint field and saved pending candidate is preserved exactly. Training hours add the recorded active intervals before and after restart; the offline gap is excluded from compute time, but remains part of the expired attempt's wall-clock history. Model-execution elapsed time refers to the recovery clock. The earlier execution allowance was not met, regardless of recovery completion. Parent deadlines remain immutable. No stopping or checkpoint selection depends on interim accuracy. The untrained baseline's older outcomes are reused with their source hash; its fresh outcomes are newly evaluated. This is a single-seed, pilot-motivated follow-up, not a test of seed robustness.

A post-restart numerical diagnostic compared 16 naturally replayed GPU responses from the unfinished candidate. Exact response, token and verifier matches: True; mismatched rows: 0. Only timestamps and batch duration were excluded. This checks the observed candidate replay, not arbitrary hardware changes or a complete optimization trajectory.

## Data, evaluation and preregistered analysis

Synthetic problems have exact rational witnesses. Global number-multiset isolation prevents permutations or alternative targets crossing splits. Only constants supplied in the prompt and allowlisted arithmetic are accepted. Greedy test accuracy at the final scheduled checkpoint is primary. The directional hypothesis is falsified for this pilot if main minus control accuracy is nonpositive. A paired problem bootstrap quantifies conditional uncertainty. Sampled pass@k follows the unbiased subset estimator [8]. Validation cannot select checkpoints. Exact dataset hashes, schedules, timing rules and sampling settings are recorded in reports/preregistration-recovery.md and configs/continuation-recovery.json.

Dataset counts: train 8192, validation 512, test 2048, hard 1024, fresh 2048. Greedy responses allow at most 128 new tokens. The sampled evaluation uses 16 responses on each of 256 fixed test problems, with temperature 1.0, top-p 1.0 and top-k disabled. pass@k = 1 - C(n-c,k)/C(n,k), where c is the number of correct samples. Intervals use 10000 paired resamples, seed 20261107.

## Results and claims

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference.

| Condition | Fresh accuracy | 95% interval | Hard accuracy | pass@1 | pass@8 | Status |
|---|---|---|---|---|---|---|
| original | 0.24% | 0.05% - 0.49% | 0.10% | 0.12% | 0.89% | complete |
| main | 7.37% | 6.25% - 8.50% | 0.39% | 2.27% | 10.66% | complete |
| reward_only | 7.71% | 6.54% - 8.89% | 0.88% | 2.47% | 11.95% | complete |

original: greedy problems fresh 2048/2048, test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. main: greedy problems fresh 2048/2048, test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. reward_only: greedy problems fresh 2048/2048, test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256.

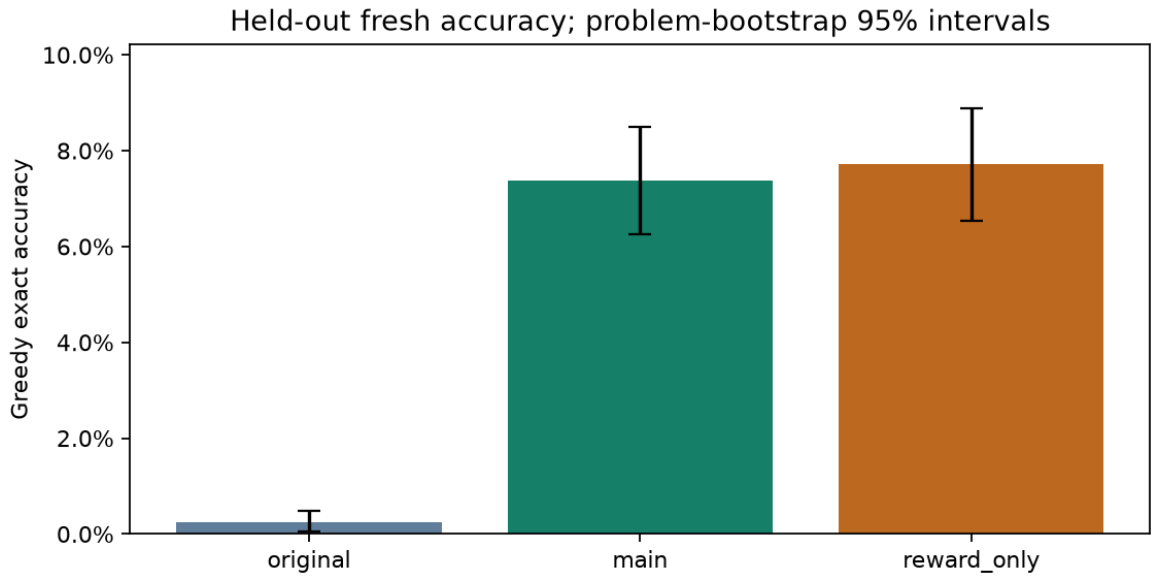
| Condition | Parent fresh accuracy | Final fresh accuracy | Added generations | Added train tokens | Added train hours |
|---|---|---|---|---|---|
| main | 1.17% | 7.37% | 81 | 2322597 | 5.060 |
| reward_only | 1.27% | 7.71% | 81 | 2434805 | 5.138 |

Fresh accuracy is primary. The change contrasts and old-test comparison below are descriptive secondary analyses. Train tokens in the later diagnostic table are cumulative; training hours and this table's added tokens concern the continuation only.

Main minus control: -0.342 percentage points; paired 95% interval [-1.025, 0.342], on 2048 paired problems.

| Comparison | Difference (pp) | Paired 95% interval (pp) | Problems |
|---|---|---|---|
| Main - reward-only (primary) | -0.342 | [-1.025, 0.342] | 2048 |
| main - original (secondary) | 7.129 | [5.957, 8.301] | 2048 |
| reward_only - original (secondary) | 7.471 | [6.299, 8.691] | 2048 |
| main: final - parent (secondary) | 6.201 | [4.932, 7.422] | 2048 |
| reward_only: final - parent (secondary) | 6.445 | [5.176, 7.715] | 2048 |
| Difference in changes (secondary) | -0.244 | [-0.977, 0.488] | 2048 |
| Old-test main - control (secondary) | -0.586 | [-1.172, -0.049] | 2048 |

Comparisons with the original checkpoint are descriptive secondary estimates with unadjusted intervals.




Percentile bootstrap intervals can collapse at observed boundaries or when paired outcomes are identical. A zero-width interval does not prove zero population error or equivalence on unseen problems.

## Probe performance, validity and resources

| Condition | Probe accuracy | Fresh validity | Hard validity | Train tokens | Eval tokens | Train hours |
|---|---|---|---|---|---|---|
| original | 0.00% | 29.74% | 10.64% | 0 | 481714 | 0.000 |
| main | 18.75% | 95.75% | 72.56% | 3121272 | 202961 | 5.060 |
| reward_only | 18.75% | 95.85% | 76.46% | 3249607 | 198224 | 5.138 |

## Within-problem expression diversity

| Condition | Greedy correct structures | Complete sampled problems | Mean correct structures / sampled problem |
|---|---|---|---|
| original | 5 | 256 | 0.0156 |
| main | 151 | 256 | 0.1992 |
| reward_only | 158 | 256 | 0.2188 |

Greedy structure counts equal the number of successful test problems because only one answer is generated per problem. The sampled statistic counts distinct correct canonical ASTs within each problem before averaging, including zeros. It measures syntactic diversity, not algebraic or semantic equivalence.

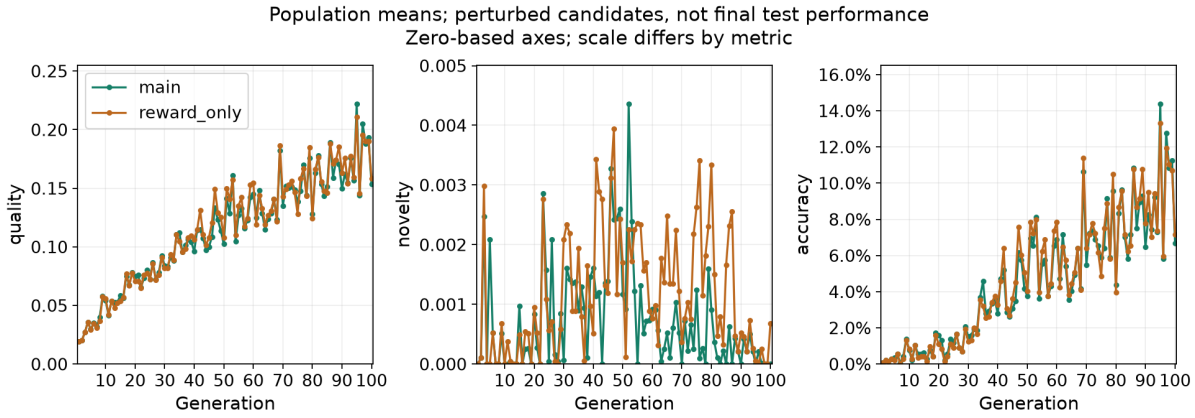
Train tokens count candidates in committed generations; evaluation tokens count persisted evaluation responses. Calibration, validation, interrupted batches and replay costs are excluded from these two table columns. Training hours are wall-clock phase duration and include checkpointing. Probes are training examples, not held-out data.

## Did the novelty bonus affect optimization?

Descriptive diagnostic: compare the main arm's Z(Q) + 1.0 N with Z(Q) on its recorded populations. The control computes novelty but applies coefficient zero, so its novelty-induced weight changes are zero. This local comparison does not simulate the counterfactual training trajectory. main: positive novelty in 225 of 3000 candidates; changed update weights in 79 of 100 generations (182 candidate weights). reward_only: positive novelty in 306 of 3000 candidates; changed update weights in 0 of 100 generations (0 candidate weights).

## Training dynamics and output inspection

Population metrics describe perturbed candidates, not the master checkpoint. Reward is the optimization signal; no cross-entropy loss is optimized in this pilot. The interactive result page shows the same first test problems across methods, including failures, without choosing examples by outcome.




## Fixed example: original

Use the numbers [44, 7, 29] exactly once each to make 80. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: (44 * (7 - 29)) / 44

## Fixed example: main

Use the numbers [44, 7, 29] exactly once each to make 80. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer>44 - (7 - 29)</answer>

## Fixed example: reward_only

Use the numbers [44, 7, 29] exactly once each to make 80. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer>44 - (7 - 29)</answer>

## Implications and discussion

Both arms improved over their earlier checkpoints when evaluated on the same fresh problems. Both trained checkpoints score higher than the original model on this test set; the main-versus-control comparison determines whether novelty adds a benefit. Both also produce more legal expressions on the three/four-number test problems. Legality alone does not mean the target was reached. This pilot is a controlled objective intervention under one initialization/seed schedule. It can identify a conditional difference or a feasibility failure, but cannot establish a general advantage of novelty training. Any improvement must be checked across additional seeds and separated from probe memorization. A rise in novelty alone, or a selected qualitative example, is insufficient evidence.

## Limitations

Training-seed sensitivity is unmeasured. Embedding similarity is a proxy and may reward irrelevant wording; canonical expression trees are an independent syntactic diagnostic, not full algebraic equivalence. Synthetic arithmetic does not measure general language quality. Similar pretraining patterns cannot be excluded. The fixed-probe objective can overfit. Small time budgets, laptop thermal variability and batch-level floating-point behavior limit extrapolation. CPU tests, report generation, audits and backups ran alongside training; elapsed times are descriptive, not isolated systems benchmarks. Secondary endpoints are descriptive; no multiple-comparison significance claims are made.

## Conclusion

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference. Multi-seed studies and archive ablations require a separately frozen amendment after this result is published.

## Appendix: reproducibility and failures

Restore the pinned interrupted evidence as described in docs/continuation.md, then run uv sync --locked, uv run pytest, uv run novelty-es download, and uv run novelty-es recover --config configs/continuation-recovery.json --run-dir PATH. uv run novelty-es report --run-dir PATH regenerates tables, figures, this paper and the static website. Repeating run resumes the same clock. Atomic checkpoints contain weights, all RNG streams, archive, cursor, config and deadline. Recorded technical failures: 0. Model-execution elapsed seconds: 19017.624. Final auditing, report preparation and publication follow model execution under the recorded allowance; expired attempts are disclosed separately. Exact exceptions and incomplete conditions are retained in the generated Markdown report and JSON artifacts.

Execution provenance: NVIDIA GeForce RTX 5070 Laptop GPU; Python 3.12.13. Initial execution revision: 93a409e91387830d24c39c919e78123e94c29790. Training implementation revision: 66884b2e7dcbd3f1f633554674dcd61faf2954e2. Frozen generations per arm: 100; inference batch: 16. The continuation timing rule uses the maximum parent generation duration plus measured checkpoint cost; it freezes the common target before continuation model execution.

Embedding revision: 1110a243fdf4706b3f48f1d95db1a4f5529b4d41. ExTra upstream revision: ea53e68036356f8f276e261e886fbbc102633be3. Complete dependency versions, dataset hashes, absolute deadline, per-phase timings, token-work accounting, raw outputs and completion flags are machine-readable in the published experiment artifacts. Full-weight noise is regenerated by a hash of perturbation seed and tensor name using an explicit PyTorch generator. Bit-level replay is scoped to the recorded hardware and software environment.

## References


1. ExTra
2. Evolution Strategies at Scale
3. Beyond the Best Guess
4. Novelty-seeking ES
5. EVOL-RL
6. DARLING
7. AC/DC
8. Evaluating Large Language Models Trained on Code

Research protocol and source


**AI statement.** Codex wrote this codebase and paper, and human cleanup and rewriting are in progress. A human provided the exact specification that was executed.